The mixed-ratio monomial asymptotic in general dimension

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2026-09-07

Abstract

Assembly of the mixed-ratio general-d monomial programme (Astra #15, units 177–181). For $h, k\colon \mathrm{Fin}(m+1) \to \mathbb{N}$ with $k_i > 0$, Mellin ratios $\ell_i = (h_i+1)/(2k_i)$, minimum λ attained on J,

$$\int_{(0,1]^{m+1}} \prod_i x_i^{h_i} e^{-\beta N \prod_i x_i^{2k_i}}\, dx \;\sim\; \frac{\Gamma(\lambda)\beta^{-\lambda}}{(|J|-1)!} \prod_{i\in J} \frac{1}{2k_i} \prod_{i\notin J} \frac{1}{h_i+1-2k_i\lambda}\; N^{-\lambda}(\log N)^{|J|-1}$$

(`monomialBoxReal_mixed_isEquivalent`; `monomialBoxReal_minRatio_isEquivalent` with $\lambda = \min_i \ell_i$ computed by `Finset.inf'`). This is the paper's Laurent coefficient $a_{-|J|}$ of eq. `a_minus_m_explicit` at cutoff $b = 1$ times the Laplace–Tauberian factor $\Gamma(\lambda)/(|J|-1)!$, now for arbitrary exponent patterns. Zero `sorry`/`axiom`.

1 The things formalised

```
def ratioExp (h k : Fin d →  ) (i : Fin d) :   := ((h i :  ) + 1) / (2 * (k i :  ))
theorem monomialBoxReal_eq_mixed (d) (h k) (hk) (  N) :
    monomialBoxReal d h k   N = (  i, 1 / (2 * k i)) * mixedBoxReal d (ratioExp h k)   N
def monomialMixedConst (h k) (l   :  ) :   :=
  Gamma l *   ^ (-l) / ((multCount (ratioExp h k) l - 1).factorial :  )
    *   i, if ratioExp h k i = l then 1 / (2 * k i) else 1 / ((h i :  ) + 1 - 2 * k i * l)
theorem monomialMixedConst_eq (hk) : monomialMixedConst h k l
    = (  i, 1 / (2 * k i)) * mixedConst (ratioExp h k) l
theorem monomialBoxReal_mixed_tendsto (d) (h k : Fin (d+1) →  ) (hk) (l  ) (hl) (h )
    (hmin :   i, l   ratioExp h k i) (hatt :   i, ratioExp h k i = l) :
    Tendsto (fun N => monomialBoxReal (d + 1) h k   N
      / (N ^ (-l) * log N ^ (multCount (ratioExp h k) l - 1))) atTop
      (  (monomialMixedConst h k l  ))
theorem monomialBoxReal_mixed_isEquivalent ... :
    (fun N => monomialBoxReal (d + 1) h k   N)
      ~[atTop] fun N => monomialMixedConst h k l   * N ^ (-l)
        * log N ^ (multCount (ratioExp h k) l - 1)
def minRatio (h k : Fin (d+1) →  ) :   :=
  Finset.univ.inf' Finset.univ_nonempty (ratioExp h k)
theorem monomialBoxReal_minRatio_isEquivalent (d) (h k) (hk) ( ) (h ) : ...
```

2 Mathematics

The coordinatewise substitution of unit 174 gives $M(N) = \prod_i \frac{1}{2k_i} W_\ell(N)$ exactly; unit 180 gives the asymptotic of W_ℓ. The two constants are identified factorwise: $\frac{1}{2k_i} \cdot \frac{1}{\ell_i - \lambda} = \frac{1}{h_i+1-2k_i\lambda}$ because $2k_i\ell_i = h_i + 1$. Positivity of the constant follows from $\Gamma(\lambda) > 0$ and $h_i + 1 - 2k_i\lambda > 0$ for $i \notin J$.

3 Paper-facing meaning

Together with unit 175 (the equal-ratio special case) this completes, for the deterministic positive-orthant unit-cutoff normal moment integral, the paper's zeta-function prediction: decay exponent equal to the minimal ratio, logarithmic degree one less than its multiplicity, and the explicit Gamma/residue coefficient — in arbitrary dimension and for arbitrary exponent patterns. Not covered: cutoffs $b \neq 1$ (which add the factors $b^{h_i+1-2k_i\lambda}$ for $i \notin J$), interior coordinates with parity factors, variable amplitudes, stochastic phase terms, and the full Taylor-tree expansion.

4 Lean notes

- A local hypothesis named `h` shadows the exponent function `h : Fin d →` and produces a baffling "argument has type Tendsto ... but is expected to have type Fin (d+1) → "; name such locals `hT`.
- Goals produced by `Eventually.of_forall fun N => ?_` arrive with unreduced `(fun N => ...) N` on the sides; run `beta_reduce` before `rw`.
- In the case split of the constant identification, the nonvanishing of $h_i + 1 - 2k_i\lambda$ is derived from $\ell_i \neq \lambda$ by `field_simp` and `linarith`, then supplied to a second `field_simp`.
- `Finset.inf'_le _ (Finset.mem_univ i)` typechecks against `univ.inf' Finset.univ_nonempty f` by proof irrelevance of the nonemptiness witness.